INCH-POUND

MS91189E
7 September 2007
SUPERSEDING
MS91189D
31 October 1972

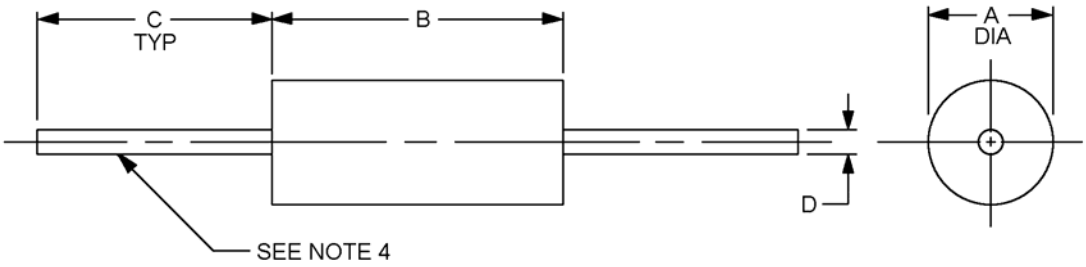
MILITARY SPECIFICATION SHEET

# COILS, RADIO FREQUENCY, MOLDED, FIXED SUBMINIATURE (PHENOLIC CORE), TYPES LT4K002 TO LT4K016, INCLUSIVE

Inactive for new design.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the products described herein shall consist of this specification and MIL-PRF-15305.




| LTR | Dimensions in inches with metric equivalents (mm) in parentheses | |
|---|---|---|
| | Minimum | Maximum |
| A | .250 (6.35) | .313 (7.95) |
| B | .875 (22.23) | 1.000 (25.40) |
| C | 1.250 (31.75) | 1.625 (41.28) |
| D | .026 (0.66) | .030 (0.76) |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. These coils are intended to be supported by their bodies.
4. Solderable/weldable lead wire, number 21 AWG.

FIGURE 1. Dimensions and configuration.



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REQUIREMENTS:

Style: LT4

Grade: 1
Class: B

Design, construction, and physical dimensions: See figure 1.

Weight: 3.9 grams, maximum.

Operating temperature range: -55°C to +125°C.

Ambient temperature: +90°C ±5°C.

Temperature rise: 35 °C, maximum.

Terminal pull: 5 pounds, minimum.

Altitude: 70,000 feet.

Shock (specified pulse): Method 213 of MIL-STD-202, test condition I, is applicable.

Dielectric withstanding voltage (sea level): Method 301 of MIL-STD-202, test voltage 1000 V rms for a minimum of 60 seconds.

Barometric pressure (reduced): Method 105 of MIL-STD-202, test condition C, test voltage of 200 V rms for a minimum of 60 seconds.

Electrical characteristics: See tables I and II.

Inductance: See table I.

Q values: See table I.

Self-resonant frequency (SRF): See table I.

DC resistance (DCR): See table I.

Part or Identifying Number (PIN): MS91189 - (dash number from table I).



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TABLE 1. Electrical characteristics (initial).

| Dash Number MS91189 1/ | Type designation | Inductance (μH) ±10% | Q (min) | Test Frequency (MHz) | SRF Minimum (MHz) | DC resistance Max. (ohms) | Rated DC current (mA) |
|---|---|---|---|---|---|---|---|
| -14 | LT4K002 | 1.2 | 60 | 7.9 | 170 | 0.075 | 2,400 |
| -15 | LT4K003 | 1.5 | 60 | 7.9 | 160 | 0.090 | 2,150 |
| -16 | LT4K004 | 1.8 | 60 | 7.9 | 140 | 0.135 | 1,750 |
| -17 | LT4K005 | 2.2 | 60 | 7.9 | 125 | 0.160 | 1,600 |
| -18 | LT4K006 | 2.7 | 60 | 7.9 | 115 | 0.220 | 1,350 |
| -19 | LT4K007 | 3.3 | 60 | 7.9 | 100 | 0.305 | 1,150 |
| -20 | LT4K008 | 3.9 | 60 | 7.9 | 95 | 0.450 | 955 |
| -21 | LT4K009 | 4.7 | 60 | 7.9 | 90 | 0.560 | 860 |
| -22 | LT4K010 | 5.6 | 60 | 7.9 | 80 | 0.745 | 745 |
| -23 | LT4K011 | 6.8 | 60 | 7.9 | 75 | 1.05 | 635 |
| -24 | LT4K012 | 8.2 | 60 | 7.9 | 68 | 1.40 | 550 |
| -25 | LT4K013 | 10.0 | 60 | 7.9 | 60 | 1.90 | 460 |
| -26 | LT4K014 | 12.0 | 40 | 2.5 | 53 | 2.65 | 395 |
| -27 | LT4K015 | 15.0 | 40 | 2.5 | 50 | 3.25 | 355 |
| -28 | LT4K016 | 18.0 | 40 | 2.5 | 45 | 4.15 | 315 |

1/ Former MS part numbers MS91189-29 thru MS91189-38 have been superseded by MS75103-1 thru MS75103-10, respectively.

TABLE II. Electrical characteristics (final).

| Inspection group | Allowable variation from Initial measurement | | Allowable percent from specified minimum value in electrical characteristics (initial) table | |
|---|---|---|---|---|
| | Inductance (percent) | DC resistance | Self-resonant frequency | Q |
| Qualification inspection | | | | |
| Group II | ±2 | --- | --- | -10 |
| Group III | ±5 | ±(3% +.001 ohm) | -8 | -10 |
| Group IV | ±5 | ±(2% +.001 ohm) | -10 | -15 |
| Conformance inspection group C | | | | |
| Subgroup I | ±2 | --- | --- | -10 |
| Subgroup II | ±5 | ±(2% +.001 ohm) | -10 | -15 |
| Subgroup III | ±5 | ±(3% +.001 ohm) | -8 | -10 |

Application notes:

1. The polarizing voltage during the moisture resistance tests is applied with the positive lead connected to the coil terminals tied together, and the negative lead connected to the metal strap.



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Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents.

MIL-PRF-15305
MIL-STD-202

Custodians:
Army – CR
Navy - EC
Air Force - 11
DLA – CC

Preparing activity:
DLA – CC

Project 5950-2007-052

Review activities:
Army – AR, MI
Navy – AS, MC, OS
Air Force – 19, 99

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